

Practical Stability of Continuous-Time Switched Systems Without a Common Equilibria and Governed by a Time-Dependent Switching Signal

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Abstract—In this paper, the problem of practical stability of some classes of continuous-time switched systems is studied. The main results of this paper include some sufficient conditions concerning practical stability of continuous-time switched non-linear systems without a common equilibria for all subsystems. In this class of switched systems, the equilibrium point varies discontinuously according to a time-dependent switching signal. So, stability with respect to a set, rather than a particular point, is discussed. Using this preliminary result, we present sufficient conditions in the form of linear matrix inequalities (LMIs) for practical stability of a particular class of switched systems without common equilibria: the switched affine systems. An illustrative example is presented to show the validity of the results.

I. INTRODUCTION

Continuous-time switched systems are typically represented by equations in the general form

$$\dot{x}(t) = f_{\sigma(\cdot)}(x(t)), \quad x(t_0) = x_0, \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the state, t_0 is the initial time and σ is a piecewise constant function called *switching signal*. This function σ can be time-dependent (when it only depends on time t), state-dependent (when it depends on the value of the state x at time t as well) or dependent on the switching history, which means that the switching exhibits some degree of memory. In this paper, we consider the switching signal is only time-dependent, so σ is a function of time defined as $\sigma(t) : I = [t_0, t_0 + T_f) \rightarrow S$, where T_f is a specified (finite) constant or $T_f = \infty$ and $S = \{1, \dots, N\}$ is a finite set of integers. Hence, given a set of vector fields (or subsystems) $\{f_1, \dots, f_N\}$, the switching signal is such that $f_{\sigma(t)} \in \{f_1, \dots, f_N\}$, for each $t \in [t_0, t_0 + T_f)$. This obviously imposes a discontinuity on $f_{\sigma(t)}$ since this vector jumps instantaneously from f_i to f_j for some $i \neq j$, $i, j \in S$, once switching occurs. The instants of time at which $f_{\sigma(t)}$

is discontinuous, i.e., $t_1, t_2, \dots, t_k, \dots \in I$ ($t_0 < t_1 < t_2 < \dots < t_k < \dots$), are called *switching times*. We also say that subsystem $i_k \in S$ is *active* in the interval $[t_k, t_{k+1})$ if $\sigma(t) = i_k$ for $t \in [t_k, t_{k+1})$.

The analysis of the dynamic behaviour of continuous-time switched systems has been addressed by several authors, whose studies are typically focused on stability [1], [2], [3], controllability, observability [4], [5] and the design of controllers with guaranteed stability and performance [6], [7]. In general, the main results in these areas assume that all subsystems of (1) share a common equilibrium (typically the origin $x = 0$, i.e., $f_i(0) = 0$ for all $i = 1, \dots, N$), and hence the stability of such systems is actually the stability of the common equilibrium [6], [8], [9], [10].

For this class of switched systems, some results of asymptotic stability of the origin can be established from direct methods, making use of an auxiliary scalar-valued Lyapunov function. By using direct methods, the global asymptotic stability of the origin is guaranteed if there exists a common, continuously differentiable, positive-definite real-valued Lyapunov function $v(x(t))$ for all subsystems $\{f_1, \dots, f_N\}$, such that

$$\frac{\partial v}{\partial x} f_i < -\lambda_i v, \quad \lambda_i > 0, \quad \forall i = 1, \dots, N, \quad x \neq 0. \quad (2)$$

The existence of a common Lyapunov function for all subsystems $\{f_1, \dots, f_N\}$ ensures that the equilibrium point $x = 0$ is globally asymptotically stable for any arbitrary switching signal [6]. Unfortunately, it is well known that, depending on the system behaviour and its dimensions, the search for a feasible solution for inequality (2) can be a very difficult task or a solution may not even exist, once that some switched systems may be stable only under restricted switching signals. To study the stability of switched systems under restricted switching rules, the necessity of a common Lyapunov function for all subsystems can be replaced by the existence of a family of continuously differentiable, radially unbounded, positive-definite real-valued Lyapunov functions $\{V_1(x(t)), \dots, V_N(x(t))\}$ such that

$$\frac{\partial V_i}{\partial x} f_i < -\lambda_i V_i, \quad \lambda_i > 0, \quad \forall i = 1, \dots, N, \quad x \neq 0. \quad (3)$$

and,

$$V_{i_{k+1}}(x(t_k)) \leq V_{i_k}(x(t_k)), \quad (4)$$

for every switching time $t_k \in I$ at which σ switches from i_k to i_{k+1} , where $i_k, i_{k+1} \in S$, $i_k \neq i_{k+1}$ [1], [3]. In spite of

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condition (2), this approach allows to the Lyapunov function $v(x(t)) = V_{\sigma(t)}(x(t))$ to have certain discontinuities (at the switching times), which turns out to be attractive for stability analysis of switched systems. The asymptotic stability of the origin is then verified when the subsystems $\{f_1, \dots, f_N\}$ are individually stable (condition (3)) and $v(x(t))$ is uniformly decreasing for all $t \in I$ (condition (4)).

A less conservative result (in comparison with both conditions (2) and the pair (3)-(4)) is given in [2], [6], where the above nonincreasing condition on the Lyapunov functions is relaxed. Basically, it is replaced by a weaker condition which imposes that the sequence $v(x(t_k)), t_0, t_1, \dots, t_k, \dots \in I$ must converge uniformly to zero, where $v(x(t_k)) = V_{i_k}(x(t_k))$ when σ switches to mode i_k at the switching time t_k . In other words, it is required that

$$V_{i_{k+1}}(x(t_{k+1})) \leq V_{i_k}(x(t_k)), \quad i_k, i_{k+1} \in S, \quad i_k \neq i_{k+1}, \quad (5)$$

for all successive switching times k and $k+1$, where $\sigma(t) = i_k \in S, \forall t \in [t_k, t_{k+1})$.

On the other hand, when there is not a common equilibria for all subsystems of (1), the stability analysis becomes more complicated. In this case, asymptotic stability is usually investigated in the sense of searching for a certain switching signal that drives the system trajectories to a particular equilibrium point. Hence, the main results proposed in this area consider that the switching signal σ is state-dependent and/or dependent on the switching history [11], [12], [13].

Interesting approaches concerning stability of switched systems without a common equilibria were investigated in [14], [15], [16], [17] in terms of *practical stability* and in [18], where some conditions are provided in order to guarantee that the trajectories of a switched system with multiple distinct equilibria converge to a certain set and remains in it for all the time. Generally speaking, the practical stability concept deals with two sets $\Omega_1 \subset \mathbb{R}^n$ and $\Omega_2 \subset \mathbb{R}^n$ satisfying $\Omega_1 \subset \Omega_2$, which are specified for the initial state and the entire system state, respectively. These two sets do not have to include the origin and they can be specified in terms of physical limitations of the system variables. The practical stability requires that if the initial state is in Ω_1 , then the system state should stay in Ω_2 for all $t \in I$ [15]. Hence, unlike the classical stability definition which is based on the existence of Ω_1 for any Ω_2 , here both of the sets Ω_1 and Ω_2 are given and so they does not vary. Stability with respect to a set, rather than a particular point, is then the basis of the practical stability concept.

Hence, motivated by the studies addressed in [15], [16], this paper aims at providing new results on practical stability of the switched system (1) without a common equilibria and under a time-dependent switching signal. This is done by following the ideas given in [2], [6] (and already previously presented here), in terms of the calculation of a dwell-time for the switching signal σ that guarantees stability (in the sense of practical stability) to the switched system by solving a feasibility problem on a suitable set of Lyapunov inequalities.

From this general result, we focus on the development of sufficient conditions for practical stability and the calculation of a dwell-time for the important class of switched systems without common equilibria referred in the literature as *switched affine systems* [11], [17], [19], [20], [21]. The obtained results for this class of systems are formulated as linear matrix inequalities (LMIs) constraints, which can be easily solved by any interior-point algorithm [22].

This paper is organized as follows. Section 2 provides the definitions of practical stability and the problem formulation. In section 3, we present the main result on practical stability for the switched system (1) without a common equilibria and considering a time-dependent switching control. Using the result of Section 3, the Section 4 presents sufficient conditions in the form of LMIs for practical stability of switched affine systems. Section 5 presents a illustrative result and its corresponding results, which demonstrate the effectiveness of the approach proposed in this paper, and section 6 contains the conclusions and some final remarks on the proposed approach.

II. PRELIMINARIES AND PROBLEM FORMULATION

In this section we first establish the notation used throughout the paper. $\mathbb{R}^n$ denotes the n -dimensional Euclidean space, $\mathbb{R}^{n \times m}$ is the set of $n \times m$ real matrices. For two elements a and b in $\mathbb{R}^n$, $\{a, b\}$ denotes the set constituted by only these two elements, while $[a, b]$ denotes the set containing all the points in the line segment between a and b . In addition, (a, b) denotes the set containing all the points of $[a, b]$, except the point b . For matrices and vectors $()'$ means transposition. For a symmetric matrix P , $P \succ 0$ ($P \prec 0$) denotes positive (negative) definiteness. In some cases, we simplify the quadratic form $y'Wy$ by $y'W(*)$. Positive (negative) semi-definiteness is denoted by $P \succeq 0$ ($P \preceq 0$). For a set $\Omega \subset \mathbb{R}^n$, we use $\bar{\Omega}$ and Ω^c to denote the closure and the complement of Ω , respectively.

Now, consider again the switched system (1). We do not assume the existence of a common equilibria for all subsystems $\{f_1, \dots, f_N\}$. The switching signal is defined in the form

$$\sigma(t) = i_k \in S = \{1, \dots, N\}, \quad \forall t \in [t_k, t_{k+1}), \quad (6)$$

where t_k and t_{k+1} are two consecutive switching times that satisfies

$$t_{k+1} - t_k \geq T, \quad (7)$$

for all the switching times $t_1, t_2, \dots, t_k, t_{k+1}, \dots \in I$ and the index $i_k \in S$ is arbitrarily selected at each of these switching times. Also, T is a positive number denoted in the literature as being a dwell-time of the switching signal $\sigma(t)$.

Definition 1 (Dwell-time [3]): A positive number T is called a *dwell-time* of the switching signal $\sigma(t)$ if the time interval between any two consecutive switchings k and $k+1$ is no smaller than T .

Since we are not assuming a common equilibria for all subsystems and the switching signal $\sigma(t)$ is only time-dependent, it is not possible to guarantee an asymptotic convergence of the system trajectories to a specific equilibrium point. Hence, stability with respect to a set, rather than a particular point, is studied in this paper in terms of practical stability.

Definition 2 (Practical stability [15]): The switched system (1) is *practically stable* with respect to $(\Omega_1, \Omega_2, t_0, T_f)$, where Ω_1 and Ω_2 are two non-empty subsets of $\mathbb{R}^n$ satisfying $\Omega_1 \subset \Omega_2$, if $x(t_0) \in \Omega_1$ implies $x(t) \in \Omega_2$, $\forall t \in I = [t_0, t_0 + T_f)$.

Definition 3 (Practical instability [15]): The switched system (1) is *practically unstable* with respect to $(\Omega_1, \Omega_2, t_0, T_f)$, where Ω_1 and Ω_2 are two non-empty subsets of $\mathbb{R}^n$ satisfying $\Omega_1 \subset \Omega_2$, if it is not practically stable with respect to $(\Omega_1, \Omega_2, t_0, T_f)$.

The first problem under consideration in this paper can be stated as follows.

Problem 1: Given Ω_1 , Ω_2 , t_0 and T_f , determine a dwell-time $T > 0$ such that the switched system (1), without a common equilibria, is practically stable with respect to $(\Omega_1, \Omega_2, t_0, T_f)$, for every switching signal $\sigma(t)$ satisfying (6)-(7).

Remark 1: Set Ω_2 is a representation of an operating security region of the system, while set Ω_1 involves all the initial conditions reflecting the perturbations at which the system is subject during its operation.

The result of practical stability established for Problem 1 is based on direct methods, using the piecewise Lyapunov function $v(x(t)) = V_{\sigma(t)}(x(t))$, which is composed by a family of continuously differentiable, radially unbounded, real-valued functions $\{V_1(x(t)), \dots, V_N(x(t))\}$, each one associated with its correspondent subsystem of the set $\{f_1(x(t)), \dots, f_N(x(t))\}$. So, $v(x(t))$ is continuous for all $t \in I$, except at the switching times $t_1, t_2, \dots, t_k, \dots \in I$. Due to these discontinuities, some restrictions are imposed on switching to guarantee stability to the system.

From the solution of the Problem 1, we derive some results on practical stability for a particular class of switched systems without a common equilibria, i.e., the switched affine systems described as

$$\dot{x}(t) = f_{\sigma(t)}(x(t)) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, \quad x(t_0) = x_0, \quad (8)$$

with a switching signal $\sigma(t)$ given by (6)-(7). So we have that, for each $t \in I$, the switching signal $\sigma(t)$ is such that $f_{\sigma(t)}(x(t)) \in \{f_1(x(t)), \dots, f_N(x(t))\}$, where $f_i(x(t)) = A_i x(t) + b_i$, $A_i \in \mathbb{R}^{n \times n}$, $b_i \in \mathbb{R}^n$ and $i = 1, \dots, N$. Assuming that all matrices A_i are nonsingular, each subsystem has a single equilibrium point at $x_{e_i} = -A_i^{-1}b_i$, $i = 1, \dots, N$.

Once we have introduced the switched affine system (8), the second problem under consideration can be stated as follows.

Problem 2: Given Ω_1 , Ω_2 , t_0 and T_f , determine a dwell-time $T > 0$ such that the switched affine system (8) is

practically stable with respect to $(\Omega_1, \Omega_2, t_0, T_f)$, for every switching signal $\sigma(t)$ satisfying (6)-(7).

Remark 2: The solution of Problem 2 is formulated as a search for continuous piecewise quadratic Lyapunov functions satisfying some LMI constraints.

III. PRACTICAL STABILITY OF SWITCHED NONLINEAR SYSTEMS WITHOUT A COMMON EQUILIBRIA

This section is dedicated to provide sufficient conditions that guarantee practical stability of the switched system (1) with respect to $(\Omega_1, \Omega_2, t_0, T_f)$ for every switching signal $\sigma(t)$ satisfying (6)-(7).

The next theorem provides the theoretical basis toward a possible solution of this problem (the problem 1 presented in the last section).

Theorem 1: Let Ω_1 , Ω_2 , t_0 and a finite constant T_f (or $T_f = \infty$) be given (with $\Omega_1 \subset \Omega_2$). The switched system (1) is practically stable with respect to $(\Omega_1, \Omega_2, t_0, T_f)$ for every switching signal $\sigma(t)$ satisfying (6)-(7), if there exist a dwell-time $T > 0$, a family of real-valued functions $\{V_1, \dots, V_N\}$, which are all differentiable, positive definite, radially unbounded, in Ω_2 and a positive number $0 < \mu \leq 1$ such that

- (i) $\dot{V}_i(x(t)) \leq 0, \forall i \in S, \forall t \in I, \forall x \in \Omega_2$,
- (ii) $V_{i_{k+1}}(x(t_{k+1})) \leq \mu V_{i_k}(x(t_{k+1} - T))$,
 $\forall i_k, i_{k+1} \in S, i_k \neq i_{k+1}, \forall t \in I, \forall x(t_{k+1}) \in \Omega_2$.

Proof: The proof is by contradiction. Let $x(t)$ be a solution of (1) with $x(t_0) \in \Omega_1$. Let us assume that there exists a $\bar{t} \in I = [t_0, t_0 + T_f)$ such that $x(\bar{t}) \notin \Omega_2$ for the first time. Let $t_1, t_2, \dots, t_m$ denote the switching times before $\bar{t}$. From (6)-(7), we have that $\sigma(t) = i_k \in S = \{1, \dots, N\}$, $\forall t \in [t_k, t_{k+1})$, $k = 0, 1, \dots, m-1$, where $t_{k+1} = t_k + T_k$ with $T_k \geq T > 0$.

Let us consider the real-valued Lyapunov function $v(x(t)) = V_{\sigma(t)}(x(t))$, which is sharing the same switching signal $\sigma(t)$ of the switched system (1). This means that there exists a set of real-valued functions $\{V_1(x(t)), \dots, V_N(x(t))\}$, each one associated to its correspondent subsystem of the set $\{f_1(x(t)), \dots, f_N(x(t))\}$. Since $V_i(x(t))$, $i = 1, \dots, N$, satisfies local Lipschitz conditions, we have

- $\sigma(t) = i_m \in S, \forall t \in [t_m, \bar{t}]$,

$$v(x(\bar{t})) = V_{i_m}(x(\bar{t})) = V_{i_m}(x(t_m)) + \int_{t_m}^{\bar{t}} \dot{V}_{i_m}(x(\tau)) d\tau$$

- $\sigma(t) = i_{m-1} \in S, \forall t \in [t_{m-1}, t_m)$,

$$v(x(t_m^-)) = V_{i_{m-1}}(x(t_m^-)) = V_{i_{m-1}}(x(t_{m-1})) + \int_{t_{m-1}}^{t_m} \dot{V}_{i_{m-1}}(x(\tau)) d\tau$$

$$\begin{aligned}
& \bullet \sigma(t) = i_{m-2} \in S, \quad \forall t \in [t_{m-2}, t_{m-1}), \\
& v(x(t_{m-1}^-)) = V_{i_{m-2}}(x(t_{m-1}^-)) = V_{i_{m-2}}(x(t_{m-2})) + \\
& \quad \int_{t_{m-2}}^{t_{m-1}} \dot{V}_{i_{m-2}}(x(\tau)) d\tau \\
& \quad \vdots \\
& \bullet \sigma(t) = i_0 \in S, \quad \forall t \in [t_0, t_1), \\
& v(x(t_1^-)) = V_{i_0}(x(t_1^-)) = V_{i_0}(x(t_0)) + \int_{t_0}^{t_1} \dot{V}_{i_0}(x(\tau)) d\tau
\end{aligned}$$

Using the inequality (i) of Theorem 1 in the above sequence, we have

$$v(x(\bar{t})) = V_{i_m}(x(\bar{t})) \leq V_{i_m}(x(t_m)) \quad (9)$$

$$V_{i_{m-1}}(x(t_m^-)) \leq V_{i_{m-1}}(x(t_{m-1})) \quad (10)$$

$$V_{i_{m-2}}(x(t_{m-1}^-)) \leq V_{i_{m-2}}(x(t_{m-2})) \quad (11)$$

$$\vdots$$

$$V_{i_0}(x(t_1^-)) \leq V_{i_0}(x(t_0)) = v(x(t_0)) \quad (12)$$

Now, we show that for any two consecutive switching times t_k and t_{k+1} it is true that $v(x(t_{k+1})) \leq \mu v(x(t_k))$. For that, using the inequality (ii) of Theorem 1, we have

$$\begin{aligned}
v(x(t_{k+1})) &= V_{i_{k+1}}(x(t_{k+1})) \\
&\leq \mu V_{i_k}(x(t_k + T_k - T)) \\
&\leq \mu \left(V_{i_k}(x(t_k)) + \int_{t_k}^{t_k + T_k - T} \dot{V}_{i_k}(x(\tau)) d\tau \right) \\
&\leq \mu V_{i_k}(x(t_k)) = \mu v(x(t_k)), \quad (13)
\end{aligned}$$

where the second inequality holds once that for every $T_k - T \geq 0$ it is true that $\dot{V}_{i_k}(x(t)) \leq 0$.

Now, using inequalities (13) and (9)-(12) it follows that

$$\begin{aligned}
v(x(\bar{t})) &\leq V_{i_m}(x(t_m)) \leq \mu V_{i_{m-1}}(x(t_{m-1})) \leq \\
&\leq \mu^2 V_{i_{m-2}}(x(t_{m-2})) \leq \dots \leq \mu^{N(t_0, \bar{t})} V_{i_0}(x(t_0)) = \\
&= \mu^{N(t_0, \bar{t})} v(x(t_0)), \quad (14)
\end{aligned}$$

where $N(a, b)$ denotes the number of switching times on the time interval $[a, b)$.

Once that $0 < \mu \leq 1$ we have from (14) that

$$v(x(\bar{t})) \leq v(x(t_0)) \leq \max_i \sup_{x \in \Omega_1} V_i(x(t_0)), \quad (15)$$

which implies that $x(\bar{t}) \notin \Omega_2^c$. So, there is not a $\bar{t} \in I$ such that $x(\bar{t}) \in \Omega_2^c$, which means that $x(t) \in \Omega_2, \forall t \in I$. ■

The imposition that μ must be a positive constant equal or smaller than 1 can be very restrictive. This is because it requires that the sequence $v(x(t_0)), v(x(t_1)), \dots, v(x(t_k)), \dots$ be nonincreasing for all $t \in I$. Next theorem provides a result in which μ must be greater than 1. Hence, the above sequence is not required to be nonincreasing for all $t \in I$. This result is only valid for the cases where T_f is a finite constant.

Theorem 2: Let Ω_1, Ω_2, t_0 and a finite constant T_f be given (with $\Omega_1 \subset \Omega_2$). The switched system (1) is practically stable with respect to $(\Omega_1, \Omega_2, t_0, T_f)$ for every switching signal $\sigma(t)$ satisfying (6)-(7), if there exist a dwell-time $T > 0$, a family of real-valued functions $\{V_1, \dots, V_N\}$, which are all differentiable, positive definite, radially unbounded, in Ω_2 and a positive number $\mu > 1$ such that conditions (i) and (ii) of Theorem 1 are satisfied and

$$\begin{aligned}
& (iii) \mu^{\bar{N}(t_0, T_f)} \max_{i \in S} \sup_{x \in \Omega_1} V_i(x(t_0)) < \min_{i \in S} \inf_{x \in \Omega_2^c} V_i(x(t)), \\
& \quad \forall t \in I,
\end{aligned}$$

where $\bar{N}(t_0, T_f)$ is the maximum number of switching times on the time interval $[t_0, T_f)$.

Proof: Let us consider the maximum number of switching times $\bar{N}(t_0, T_f)$ on the time interval $[t_0, T_f)$ (which satisfies $\bar{N}(t_0, T_f) \geq N(t_0, \bar{t})$). Hence, it follows from (14) and condition (iii) that

$$\begin{aligned}
v(x(\bar{t})) &\leq \mu^{N(t_0, \bar{t})} v(x(t_0)) \leq \mu^{N(t_0, \bar{t})} \max_{i \in S} \sup_{x \in \Omega_1} V_i(x(t_0)) \\
&\leq \mu^{\bar{N}(t_0, T_f)} \max_{i \in S} \sup_{x \in \Omega_1} V_i(x(t_0)) < \min_{i \in S} \inf_{x \in \Omega_2^c} V_i(x(\bar{t})),
\end{aligned}$$

for a $\mu > 1$. So, we have that $v(x(\bar{t})) < \min_i \inf_{x \in \Omega_2^c} V_i(x(\bar{t}))$, which implies that $x(\bar{t}) \notin \Omega_2^c$. This leads to a contradiction to the original assumption. So, there is not a $\bar{t} \in [t_0, t_0 + T_f)$ such that $x(\bar{t}) \in \Omega_2^c$, which means that $x(t) \in \Omega_2, \forall t \in [t_0, t_0 + T_f)$. ■

For many physical switched systems, finding a family of functions $V_i(x(t))$ satisfying the conditions of Theorems 1 and 2 may be difficult, once that the constraints (i)-(ii) must be checked along the time interval $[t_0, t_0 + T_f)$ for all the Lyapunov functions of the set $\{V_1, \dots, V_N\}$. However, for some special classes of switched systems, as it is shown in sequence for the class of switched affine systems, more tractable results can be obtained from the general results presented in Theorems 1 and 2.

IV. PRACTICAL STABILITY OF SWITCHED AFFINE SYSTEMS

This section is dedicated to provide sufficient conditions that guarantee practical stability of the switched affine system (8) with respect to $(\Omega_1, \Omega_2, t_0, T_f)$ for every switching signal $\sigma(t)$ satisfying (6)-(7). We adopt the sets Ω_1 and Ω_2 defined as

$$\Omega_1 = \bigcap_{i=1}^N \Omega_{1_i}, \quad (16)$$

$$\Omega_2 = \bigcup_{i=1}^N \Omega_{2_i}, \quad (17)$$

where

$$\Omega_{1_i} = \{x \in \mathbb{R}^n : (x - x_{e_i})' P_i (x - x_{e_i}) + d_i < \alpha\},$$

$$\Omega_{2_i} = \{x \in \mathbb{R}^n : (x - x_{e_i})' P_i (x - x_{e_i}) + d_i < \beta\},$$

being α and β two positive scalars satisfying $\alpha < \beta$, $P_i \in \mathbb{R}^{n \times n}$ a positive definite matrix, d_i a nonnegative scalar,

$x_{e_i} = -A_i^{-1}b_i$ the equilibrium point of the subsystem i , considering that A_i is nonsingular and $i = 1, \dots, N$.

Next Theorem provides a sufficient condition for practical stability of the switched affine system (8) (Problem 2 presented in section 2). The proposed result comes from the application of Theorem 1 to the problem under consideration here by assuming a piecewise Lyapunov function given by

$$v(x(t)) = V_{\sigma(t)}(x(t)) = (x - x_{e_{\sigma(t)}})' P_{\sigma(t)} (x - x_{e_{\sigma(t)}}) + d_{\sigma(t)}, \quad (18)$$

which is composed of a family of continuously differentiable, radially unbounded, real-valued functions in the set $\{V_1(x(t)), \dots, V_N(x(t))\}$, each one associated with its correspondent subsystem of the set $\{f_1(x(t)), \dots, f_N(x(t))\}$, where $V_i(x(t)) = (x - x_{e_i})' P_i (x - x_{e_i}) + d_i$ and $f_i(x(t)) = A_i x(t) + b_i$, $i = 1, \dots, N$.

Theorem 3: Let α, β, t_0 and T_f be given with $\alpha < \beta$. Let Ω_1 and Ω_2 be the sets given by (16) and (17), respectively. The switched affine system (8) is practically stable with respect to $(\Omega_1, \Omega_2, t_0, T_f)$, $\Omega_1 \subset \Omega_2$, for every switching signal $\sigma(t)$ satisfying (6)-(7), if there exist a dwell-time $T > 0$, positive definite matrices $P_1, \dots, P_N$, nonnegative scalars $d_1, \dots, d_N < \alpha$ and a positive number μ such that the following matrix inequalities are satisfied

$$A_i' P_i + P_i A_i \leq 0, \quad (19)$$

$$\begin{bmatrix} e^{A_i' T} P_j e^{A_i T} - \mu P_i & e^{A_i' T} P_j \Delta x_{e_{ij}} \\ * & \Delta x_{e_{ij}}' P_j \Delta x_{e_{ij}} + d_j - \mu d_i \end{bmatrix} \leq 0, \quad (20)$$

$$\mu < \sqrt{\bar{N}(t_0, T_f) \beta / \alpha}, \quad (21)$$

for all $i, j \in S$, $i \neq j$, where $\bar{N}(t_0, T_f)$ is the maximum possible number of switching times on the time interval (t_0, T_f) and $\Delta x_{e_{ij}} = x_{e_i} - x_{e_j}$.

Proof: Consider, in accordance to (6)-(7), that $\sigma(t) = i \in S$, $\forall t \in [t_k, t_{k+1})$, where $t_{k+1} = t_k + T_k$ with $T_k \geq T > 0$, and that at $t = t_{k+1}$ the switching signal jumps to $\sigma(t) = j \in S$. From condition (19), it is seen that for all $t \in [t_k, t_{k+1})$, the time derivative of the Lyapunov function (18) along an arbitrary trajectory of the switched affine system (8) holds

$$\begin{aligned} \dot{v}(x(t)) &= (x(t) - x_{e_i})' (A_i' P_i + P_i A_i) (x(t) - x_{e_i}), \\ &\leq 0, \end{aligned}$$

for all $x \in \Omega_2$, which satisfies the condition (i) of Theorem 1. Now, let us rewrite condition (20) as

$$\begin{bmatrix} e^{A_i' T} P_j e^{A_i T} & e^{A_i' T} P_j \Delta x_{e_{ij}} \\ * & \Delta x_{e_{ij}}' P_j \Delta x_{e_{ij}} + d_j \end{bmatrix} \leq \mu \begin{bmatrix} P_i & 0 \\ * & d_i \end{bmatrix}, \quad (22)$$

Multiplying (22) on the right and the left by $\begin{bmatrix} e^{A_i'(T_k - T)} & 1 \end{bmatrix}'$ and $\begin{bmatrix} e^{A_i'(T_k - T)} & 1 \end{bmatrix}$, respectively, we have that

$$\begin{aligned} &\begin{bmatrix} e^{A_i' T_k} P_j e^{A_i T_k} & e^{A_i' T_k} P_j \Delta x_{e_{ij}} \\ * & \Delta x_{e_{ij}}' P_j \Delta x_{e_{ij}} + d_j \end{bmatrix} \\ &\leq \mu \begin{bmatrix} e^{A_i'(T_k - T)} P_i e^{A_i(T_k - T)} & 0 \\ * & d_i \end{bmatrix}. \end{aligned} \quad (23)$$

Using the inequality (23) we have

$$\begin{aligned} V_j(x(t_{k+1})) &= \\ &= (x(t_{k+1}) - x_{e_j})' P_j (*) + d_j \\ &= (e^{A_i T_k} (x(t_k) - x_{e_i}) + x_{e_i} - x_{e_j})' P_j (*) + d_j \\ &= (e^{A_i T_k} (x(t_k) - x_{e_i}) + \Delta x_{e_{ij}})' P_j (*) + d_j \\ &= \begin{bmatrix} x(t_k) - x_{e_i} \\ 1 \end{bmatrix}' \begin{bmatrix} e^{A_i' T_k} P_j e^{A_i T_k} & e^{A_i' T_k} P_j \Delta x_{e_{ij}} \\ * & \Delta x_{e_{ij}}' P_j \Delta x_{e_{ij}} + d_j \end{bmatrix} (*) \\ &\leq \mu \begin{bmatrix} x(t_k) - x_{e_i} \\ 1 \end{bmatrix}' \begin{bmatrix} e^{A_i'(T_k - T)} P_i e^{A_i(T_k - T)} & 0 \\ * & d_i \end{bmatrix} (*) \\ &\leq \mu \left((e^{A_i(T_k - T)} (x(t_k) - x_{e_i}) + x_{e_i} - x_{e_j})' P_i (*) + d_i \right) \\ &\leq \mu ((x(t_{k+1} - T) - x_{e_i})' P_i (*) + d_i) \\ &\leq \mu V_i(x(t_{k+1} - T)), \end{aligned}$$

which leads to the condition (ii) of Theorem 1. Now, from the sets Ω_1 and Ω_2 given by (16) and (17), respectively, we have that

$$\max_{i \in S} \sup_{x \in \Omega_1} V_i(x(t_0)) = \alpha, \quad \min_{i \in S} \inf_{x \in \Omega_2} V_i(x(t)) = \beta, \quad (24)$$

so, the constraint (21) leads to the condition (iii) of Theorem 1. ■

Remark 3: The constraint (19) imposes that all the matrices $A_1, \dots, A_N$ be stable in the usual sense of Lyapunov stability concept.

Remark 4: Once that α and β are two pre-specified parameters, the size of the sets Ω_1 and Ω_2 can be varied by manipulating the size of the scalar variables d_i and also the size of the trace of the matrix variables P_i ($i = 1, \dots, N$). This can be easily done by including additional constraints to the set of matrix inequalities of Lemma 1.

Remark 5: The constraint (20) is not an LMI, once there exist nonlinearity in the variable T , cross-products between the scalar variables μ and d_i , as well as μ and the matrix variable P_i . These nonlinearities can be eliminated by pre-specifying a value for T and also a value for μ that satisfies the condition (21). Doing so, the resulting problem is formulated as a set of LMIs given by (19) and (20) in the variables P_i and d_i , $i = 1, \dots, N$.

Next section provides an example to illustrate the above Theorem.

V. ILLUSTRATIVE EXAMPLE

Consider the switched affine system described by

$$\dot{x}(t) = A_{\sigma(t)} x(t) + b_{\sigma(t)}, \quad (25)$$

with a switching signal $\sigma(t)$ given by (6)-(7), where $S = 1, 2, 3$. We also have $x(t) \in \mathbb{R}^2$ and the time interval of interest is given by $I = [0, T_f]$, where $T_f = 50s$. The

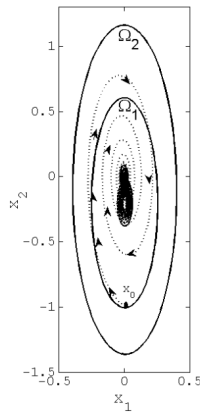


Fig. 1. System trajectory for an initial condition $x_0 \in \Omega_1$.

matrices A_i and vectors b_i , $i = 1, 2, 3$, are given by

$$\begin{aligned} A_1 &= \begin{bmatrix} -0.1 & 10 \\ -100 & -5 \end{bmatrix}, A_2 = \begin{bmatrix} -0.3 & 10 \\ -100 & -3 \end{bmatrix}, \\ A_3 &= \begin{bmatrix} -0.5 & 10 \\ -100 & -1 \end{bmatrix}, \\ b_1 &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, b_3 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}. \end{aligned}$$

The equilibrium points of subsystems 1, 2 and 3 of (25) are, respectively, $x_{e1} = [0 \ 0]'$, $x_{e2} = [0.003 \ -0.1]'$ and $x_{e3} = [0.002 \ -0.2]'$. We deal with the sets Ω_1 and Ω_2 given by (16) and (17), respectively, where the parameters α and β were pre-specified as 0.8 and 1, respectively. For the time interval of interest $I = [0, 50)$, we solved the LMI set of Theorem 3 for a dwell-time $T = 1s$ and $\mu = 1.001$, so $\bar{N}(0, 50) = 50$. The system trajectory for an initial condition $x_0 \in \Omega_1$ is shown in Figure 1. It can be seen that the trajectory remains confined in Ω_2 for all $t \in I$.

VI. CONCLUSIONS

This paper presented new results on practical stability analysis of some classes of continuous-time switched systems. Our results provide estimates of system trajectory areas, as illustrated in the example of Section 5. The motivation for focusing on practical stability of switched affine systems under a time-dependent switching signal comes from problems in the area of small-signal stability of power systems (see, for example, [23], [24], [25]), in which the system trajectories must be confined into a pre-specified region over a wide range of operating conditions. Hence, as a next step of this research, our goal is to focus on the application of the results proposed here to the problem of small-signal stability of power systems.

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